1 · The D → E → P chain (do it in this order)

- **D** from Gauss on **free** charge only: $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,enc}$. Needs symmetry (sphere/cyl/plane).
- $\mathbf{E} = \mathbf{D}/\epsilon$ (linear) or $(\mathbf{D}-\mathbf{P})/\epsilon_0$ (frozen P).
- $\mathbf{P} = \epsilon_0\chi_e\mathbf{E} = \mathbf{D} - \epsilon_0\mathbf{E} = \frac{\chi_e}{\epsilon_r}\mathbf{D} = (1-\frac{1}{\epsilon_r})\mathbf{D}$.
- **Bound:** $\sigma_b = \mathbf{P}\cdot\hat{n}$ ($\hat{n}$ OUT of the dielectric; inner surface $\hat{n} = -\hat{r}$). $\rho_b = -\nabla\cdot\mathbf{P}$.
- Linear, homogeneous: $\rho_b = -\frac{\chi_e}{1+\chi_e}\rho_f$ → **no free inside ⇒ no** ρ_b.
- $\mathbf{V} = -\int \mathbf{E}\cdot d\mathbf{l}$ region by region from ∞ inward; keep V continuous.
- Check: total bound charge $= 0$. Next to a conductor: $\sigma_f + \sigma_b = \sigma_f/\epsilon_r$.

2 · Shortcuts that skip the chain

- **No free charge + symmetry ⇒ D = 0 ⇒ E = −P/ε₀** inside, 0 outside.
- Trap: $\nabla\times\mathbf{D} = \nabla\times\mathbf{P} \neq 0$; no symmetry ⇒ D NOT fixed by ρ_f (bar electret: $\rho_f = 0$, $\mathbf{D} \neq 0$).
- Free charge only on the **surface** of a dielectric sphere ⇒ inside D=E=P=0; dielectric does nothing.
- Fill all space with ϵ: every field from free charge ÷ ϵ_r.

3 · Divergence one-liners

$F(r)\hat{r}$: $\frac{1}{r^2}(r^2F)'\cdot F(s)\hat{s}$: $\frac{1}{s}(sF)'\cdot F(z)\hat{z}$: F'. Curl of these = 0.

$P = kr^n\hat{r}$: $\rho_b = -(n+2)kr^{n-1}$. $P = ks^n\hat{s}$: $\rho_b = -(n+1)ks^{n-1}$. $\sigma_b = P(R)$.

$\hat{r}/r^2$ hides $4\pi\delta^3$; $\hat{s}/s$ hides $2\pi\delta^2$ (line charge).

4 · Uniform P (frozen)

- $\rho_b = 0$; $\sigma_b = \pm P$ only on faces P pierces (sphere: $P\cos\theta$).
- $\mathbf{E}_{in} = -N\mathbf{P}/\epsilon_0$: slab⊥ N=1, sphere $\frac{1}{3}$, cyl⊥ $\frac{1}{2}$, needle ∥ 0.
- Sphere outside = pure dipole $\mathbf{p} = \frac{4}{3}\pi R^3\mathbf{P}$: $V = \frac{PR^3\cos\theta}{3\epsilon_0 r^2}$.
- Shell $a<r<b$ uniform P: **E = 0 in cavity** (ball – ball).
- Offset balls $\pm\rho$: overlap $\mathbf{E} = -\rho\mathbf{d}/3\epsilon_0$ ($\mathbf{d}$: − to + center). $\rho\mathbf{d} \to \mathbf{P}$.
- Cavity in polarized slab (P⊥): $-\frac{2P}{3\epsilon_0}$; outside: dipole $-\frac{4}{3}\pi R^3 P$.

5 · Must-know P problems

- Sphere $\mathbf{P} = k\mathbf{r}$: $\sigma_b = kR$, $\rho_b = -3k$, $\mathbf{E}_{in} = -k\mathbf{r}/\epsilon_0$, out 0.
- Shell $\mathbf{P} = \frac{k}{r}\hat{r}$: $\rho_b = -\frac{k}{r^2}$, $\sigma_b(a) = -\frac{k}{a}$, $\sigma_b(b) = \frac{k}{b}$, $E = -\frac{k}{\epsilon_0 r}$ in shell.
- Cyl $\mathbf{P} = ks\hat{s}$: $\rho_b = -2k$, $\sigma_b = ka$, $E = -ks/\epsilon_0$.
- Bar electret (4.11): caps $\pm P$; long → two point charges; short → capacitor $E \approx -P/\epsilon_0$.

6 · Linear: standard free-charge setups

- Point q in ϵ: $E = \frac{q}{4\pi\epsilon r^2}$; bound at q: $-q\frac{\chi_e}{1+\chi_e}$.
- Conductor Q, radius a, shell to b: $D = \frac{Q}{4\pi r^2}$; $V(a) = kQ[\frac{1}{b} + \frac{1}{\epsilon_r}(\frac{1}{a}-\frac{1}{b})]$; $\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi a^2}$, $\sigma_b(b) = +\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi b^2}$.
- Ball of ϵ_r, uniform ρ_f: $D = \rho_f r/3$, $\sigma_b = \frac{\chi_e}{\epsilon_r}\frac{\rho_f R}{3}$.
- Nonuniform $\epsilon(x)$: D still $= \sigma$; $\rho_b = -dP/dx \neq 0$.

7 · Capacitors = springs

- **Stacked** (series): same $D = \sigma_f$; $E_i = \sigma/\epsilon_i$; $\frac{1}{C} = \sum\frac{d_i}{\epsilon_i A}$.
- **Side by side** (parallel): same $E = V/d$; $D_i = \epsilon_i E$; $C = \sum\frac{\epsilon_i A_i}{d}$.
- Half fill: stacked $\frac{2\epsilon_r}{1+\epsilon_r}C_0$, side $\frac{1+\epsilon_r}{2}C_0$.
- Coax: $\frac{C}{L} = \frac{2\pi\epsilon_0}{\sum\ln(b_i/a_i)/\epsilon_{ri}}$. Sphere: $\frac{1}{C} = \sum\frac{1}{4\pi\epsilon_i}(\frac{1}{a_i}-\frac{1}{b_i})$.
- HW 4.18 (ϵ_r 2, 1.5): $P = \sigma/2, \sigma/3$; $V = \frac{7\sigma a}{6\epsilon_0}$.

8 · Boundary conditions

- $D_1^\perp - D_2^\perp = \sigma_f$; $E^\parallel$ continuous; V continuous.
- No free: $\epsilon_1\partial_n V_1 = \epsilon_2\partial_n V_2$. $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$ (bends away from normal into high ε).

9 · Sphere / cylinder in uniform E_0

$V_{in} = Ar\cos\theta$, $V_{out} = -E_0 r\cos\theta + \frac{B\cos\theta}{r^2}$. BCs at R: V cont.; $\epsilon_r A = -E_0 - \frac{2B}{R^3}$.

$A = -\frac{3E_0}{\epsilon_r+2}$, $B = \frac{\epsilon_r-1}{\epsilon_r+2}E_0R^3$; $P = 3\epsilon_0\frac{\epsilon_r-1}{\epsilon_r+2}E_0$; $\sigma_b = P\cos\theta$; $E_r^{out}(0) = \frac{3\epsilon_r}{\epsilon_r+2}E_0$.

General: sphere $E_{in} = \frac{3\epsilon_{out}}{\epsilon_{in}+2\epsilon_{out}}E_0$; cyl⊥ $\frac{2\epsilon_{out}}{\epsilon_{in}+\epsilon_{out}}E_0$. Cavity: $\frac{3\epsilon_r}{1+2\epsilon_r}$. Conductor: $\epsilon_{in}\to\infty$.

Coated conductor a, shell to b (4.24): $V = A(r - \frac{a^3}{r^2})\cos\theta$, $A = \frac{-3E_0}{(\epsilon_r+2)+2(\epsilon_r-1)a^3/b^3}$.

Check: $\epsilon_r \to 1$ vacuum; $\epsilon_r \to \infty$ conductor.

10 · Images with a dielectric

q at height d above ϵ_r: above use $q' = -\frac{\epsilon_r-1}{\epsilon_r+1}q$ at $-d$; below use $q'' = \frac{2q}{\epsilon_r+1}$ at $+d$ (with k).

$F = \frac{kqq'}{(2d)^2}$; $\sigma_b = -\frac{1}{2\pi}\frac{\chi_e}{\chi_e+2}\frac{qd}{(s^2+d^2)^{3/2}}$; $W = \frac{1}{2}\cdot\frac{kqq'}{2d}$.

Line charge: same fraction.

Plane at V_0: add V_0. Charge Q on sphere in E_0: add $\frac{Q}{4\pi\epsilon r}$.

11 · Energy

- $W = \frac{1}{2}\int\mathbf{D}\cdot\mathbf{E}\,d\tau = \frac{1}{2}\int\frac{D^2}{\epsilon}d\tau$ (free work). $\frac{\epsilon_0}{2}\int E^2$ = all charge incl. bound. Frozen P, no free: $\frac{1}{2}\int D\cdot E = 0$!
- Fastest: $\frac{1}{2}QV$, $\frac{Q^2}{2C}$. Outside a sphere: always $\frac{Q^2}{8\pi\epsilon_0 R}$.
- Ball: $\frac{3Q^2}{20\pi\epsilon_0 R}$; ball of ϵ_r: $\frac{Q^2}{8\pi\epsilon_0 R}(1+\frac{1}{5\epsilon_r})$; 4.26: $\frac{Q^2}{8\pi\epsilon_0\epsilon_r}(\frac{1}{a}+\frac{\chi_e}{b})$.
- Fixed Q: W ÷ ϵ_r. Fixed V: W × ϵ_r. Force $F = \frac{1}{2}V^2\frac{dC}{dx}$ (toward higher C). Slab: $\frac{\epsilon_0\chi_e w V^2}{2d}$.
- Infinite line/cyl: energy/length diverges outside (ln).

12 · Dipoles & multipoles

- $Q = \sum q$, $\mathbf{p} = \sum q_i\mathbf{r}_i = \int\mathbf{r}\rho$; Q=0 ⇒ p origin-independent; shift: $\mathbf{p}' = \mathbf{p} - Q\mathbf{a}$.
- $V = k[\frac{Q}{r} + \frac{\mathbf{p}\cdot\hat{r}}{r^2} + \frac{1}{r^3}\sum q_i r_i^2 P_2(\cos\alpha_i)]$; $P_2 = \frac{3x^2-1}{2}$.
- $\mathbf{E}_{dip} = \frac{kp}{r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta})$.
- $\mathbf{N} = \mathbf{p}\times\mathbf{E}$, $U = -\mathbf{p}\cdot\mathbf{E}$, $\mathbf{F} = \nabla(\mathbf{p}\cdot\mathbf{E})$; uniform E: F = 0.
- Image of p in plane: ∥ part flips, ⊥ part kept. Rests ⊥ plane. Coaxial dipoles: $F = \frac{3p_1p_2}{2\pi\epsilon_0 r^4}$, attract.

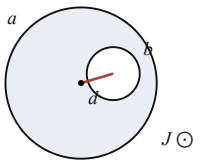
12b · Across an interface (no free charge)

$D^\perp$	continuous ($\epsilon_1E_1^\perp = \epsilon_2E_2^\perp$)
$E^\parallel$	continuous
$E^\perp$	jumps by σ_b/ϵ_0
σ_b	$\mathbf{P}_1\cdot\hat{n}_1 + \mathbf{P}_2\cdot\hat{n}_2$ (each n̂ out of its own slab)

13 · 50-minute plan

Read all (2 min) → write every standard result → easy parts first → setup + BCs = most points → 3-min stall rule → last 8 min: limits ($\epsilon_r \to 1, \infty$), units (P, D, σ: C/m²).

14 · Off-axis hole (Ampère + superposition)



- Wire radius a, hole radius b, offset $\mathbf{d}$, total current I ⇒ $J = \frac{I}{\pi(a^2-b^2)}$.
- Full wire: $I_+ = J\pi a^2 = I\frac{a^2}{a^2-b^2}$. Hole wire: $I_- = -J\pi b^2 = -I\frac{b^2}{a^2-b^2}$. ($I_+ + I_- = I$ ✓)
- Inside a uniform wire: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J}\times\mathbf{s}$ (s from ITS axis). Outside: $\frac{\mu_0 I}{2\pi s}\hat{\phi}$.
- **In hole:** $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J}\times\mathbf{d}$, uniform, $|B| = \frac{\mu_0 I d}{2\pi(a^2-b^2)}$, ⊥ to d.
- In metal: $\frac{\mu_0}{2}\mathbf{J}\times\mathbf{s} - \frac{\mu_0|I_-|}{2\pi s'}\hat{\phi}'$ (s' from hole axis).
- Outside: $\frac{\mu_0 I_+}{2\pi s}\hat{\phi} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$. On the line through both axes: add with signs.
- Opposite-J overlapping wires: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J}\times(\mathbf{c}_- - \mathbf{c}_+)$ in overlap.
- A in hole: $A_z = -\frac{\mu_0 J}{4}(s^2 - |\mathbf{s}-\mathbf{d}|^2)$ (linear in s ⇒ uniform B).

15 · Vector potential WITHOUT integrating

- $\mathbf{B} = \nabla\times\mathbf{A}$, $\nabla\cdot\mathbf{A} = 0$, $\nabla^2\mathbf{A} = -\mu_0\mathbf{J}$. **A ∥ J** (A follows the current).
- $\mathbf{B} = B(s)\hat{\phi} \Rightarrow \mathbf{A} = A_z(s)\hat{z}$, $B_\phi = -\frac{dA_z}{ds}$.
- $\mathbf{B} = B(s)\hat{z} \Rightarrow \mathbf{A} = A_\phi(s)\hat{\phi}$, $B_z = \frac{1}{s}\frac{d}{ds}(sA_\phi)$; or Stokes: $A_\phi 2\pi s = \Phi(s)$.
- $\mathbf{B} = B(z)\hat{y} \Rightarrow A_x(z)$, $B_y = \partial_z A_x$.
- Spherical $\mathbf{A} = A_\phi(r,\theta)\hat{\phi}$: $B_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\phi)$, $B_\theta = -\frac{1}{r}\partial_r(rA_\phi)$.
- Then $\mu_0\mathbf{J} = \nabla\times\mathbf{B}$: $\hat{\phi}$ field: $\frac{1}{s}(sB_\phi)'\hat{z}$; $\hat{z}$ field: $-B_z'\hat{\phi}$.
- A continuous across K; $\partial_n\mathbf{A}$ jumps by $-\mu_0\mathbf{K}$. Integral to ∞ diverges ⇒ use curl/Stokes.

uniform B	$\frac{1}{2}\mathbf{B}\times\mathbf{r} = \frac{B_0 s}{2}\hat{\phi}$
wire (out)	$-\frac{\mu_0 I}{2\pi}\ln\frac{s}{s_0}\hat{z}$
uniform-J wire (in)	$-\frac{\mu_0 J s^2}{4}\hat{z}$
solenoid	$\frac{\mu_0 nIs}{2}\hat{\phi}$ in; $\frac{\mu_0 nIR^2}{2s}\hat{\phi}$ out
sheet $K\hat{x}$	$-\frac{\mu_0 K}{2}\lvert z\rvert\hat{x}$
spin shell	in $\frac{\mu_0 R\omega\sigma}{3}r\sin\theta\hat{\phi}$; out $\frac{\mu_0 R^4\omega\sigma}{3}\frac{\sin\theta}{r^2}\hat{\phi}$
dipole	$\frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{r}}{r^2} = \frac{\mu_0 m\sin\theta}{4\pi r^2}\hat{\phi}$
finite wire	$\frac{\mu_0 I}{4\pi}\ln\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}\hat{z}$
$A = k\hat{\phi}$	$B = \frac{k}{s}\hat{z}$, $J = \frac{k}{\mu_0 s^2}\hat{\phi}$

16 · Ampère: the 4 symmetric shapes

- Wire (uniform): $\frac{\mu_0 Is}{2\pi a^2}$ in, $\frac{\mu_0 I}{2\pi s}$ out. $J\propto s^k$: $B_{in} = \frac{\mu_0 I}{2\pi s}\left(\frac{s}{a}\right)^{k+2}$.
- Sheet: $\frac{\mu_0 K}{2}$, dir $\mathbf{K}\times\hat{n}$. Two opposite sheets: $\mu_0 K$ between, 0 out.
- Slab $|z| < a$, $J\hat{x}$: $-\mu_0 Jz\hat{y}$ in, $\mp\mu_0 Ja\hat{y}$ out.
- Solenoid $\mu_0 nI$ (0 out); toroid $\frac{\mu_0 NI}{2\pi s}$; coax $\frac{\mu_0 I}{2\pi s}$ between, 0 out.
- Coaxial solenoids opposite: $\mu_0(n_1-n_2)I$, $\mu_0 n_2 I$, 0. Moving capacitor: $\mu_0\sigma v$; balance at $v = c$.
- Force on a sheet: use OTHER sheet's field (avg).

17 · Biot–Savart pieces

- Segment: $\frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$ (θ from ⊥ foot). On its line: 0.
- Semi-∞ beside end: $\frac{\mu_0 I}{4\pi s}$. Arc φ at center: $\frac{\mu_0 I\phi}{4\pi R}$.
- Loop axis: $\frac{\mu_0 IR^2}{2(R^2+z^2)^{3/2}}$. n-gon: $\frac{n\mu_0 I}{2\pi R}\sin\frac{\pi}{n}$; square $\frac{\sqrt{2}\mu_0 I}{\pi R}$.
- Finite solenoid axis $\frac{\mu_0 nI}{2}(\cos\theta_2 - \cos\theta_1)$; mouth $\frac{\mu_0 nI}{2}$. Helmholtz d=R: $\frac{8\mu_0 I}{5\sqrt{5}R}$.
- $\mathbf{F} = I\int d\mathbf{l}\times\mathbf{B}$; uniform B: closed loop F=0, curved wire = chord. Wires: $\frac{\mu_0 I_1 I_2}{2\pi d}$, same dir attract.
- Loop near wire: $\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$. $\mathbf{B} = kz\hat{x}$ loop: $Ika^2\hat{z}$.

18 · Spinning charge & dipole m

- $K = \sigma\omega s$ (disk), $\sigma\omega R\sin\theta$ (shell); $J = \rho\omega r\sin\theta$ (ball). sinθ!
- $m = \frac{Q}{2M}L$: ring $\frac{1}{2}Q\omega R^2$, disk $\frac{1}{4}$, shell $\frac{1}{3}$, ball $\frac{1}{5}$.
- Center B ($\frac{\mu_0 Q\omega}{\pi R}$): ring ¼, disk ½, shell ⅙, ball ¼. Shell inside uniform $\frac{2}{3}\mu_0\sigma\omega R$.
- $\mathbf{m} = I\mathbf{a}$; $\mathbf{B}_{dip} = \frac{\mu_0 m}{4\pi r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta})$; no monopole term.
- $\mathbf{N} = \mathbf{m}\times\mathbf{B}$, $U = -\mathbf{m}\cdot\mathbf{B}$, $\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B})$; coaxial $\frac{3\mu_0 m_1 m_2}{2\pi r^4}$.
- B_{avg} in sphere $= \frac{\mu_0}{4\pi}\frac{2\mathbf{m}}{R^3}$; solid spin ball: $\frac{\mu_0 Q\omega}{10\pi R}$.

19 · Magnetic BCs

$B^\perp$ continuous; $\mathbf{B}_a - \mathbf{B}_b = \mu_0\mathbf{K}\times\hat{n}$; $\mathbf{H}_a^\parallel - \mathbf{H}_b^\parallel = \mathbf{K}_f\times\hat{n}$; $\mu_1 H_1^\perp = \mu_2 H_2^\perp$. Inside from outside: one BC at the equator.

20 · M, bound currents, H

- $\mathbf{J}_b = \nabla\times\mathbf{M}$, $\mathbf{K}_b = \mathbf{M}\times\hat{n}$ (faces ∥ M). $\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$, $\oint\mathbf{H}\cdot d\mathbf{l} = I_f$.
- **No free I + symmetry ⇒ H = 0 ⇒ B = μ₀M.** Free I: H from Ampère, $\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M})$.
- Linear: $\mathbf{M} = \chi_m\mathbf{H}$, $\mathbf{B} = \mu\mathbf{H}$, $\mathbf{J}_b = \chi_m\mathbf{J}_f$. Coax χ_m: $I_b = \chi_m I$ on inner surface.
- Uniform M: sphere $\frac{2}{3}\mu_0\mathbf{M}$ in, $\mathbf{m} = \frac{4}{3}\pi R^3\mathbf{M}$; long rod $\mu_0\mathbf{M}$; thin disk ≈0. $\mathbf{B}_{in} = \mu_0(1-N)\mathbf{M}$, $\mathbf{H}_{in} = -N\mathbf{M}$.
- $\mathbf{M} = ks^2\hat{\phi}$: $J_b = 3ks\hat{z}$, $K_b = -kR^2\hat{z}$, B=μ₀M. $\mathbf{M} = ks\hat{z}$: $J_b = -k\hat{\phi}$, $K_b = kR\hat{\phi}$, B=μ₀M.
- Para: M ∥ B, $\chi_m>0$, pulled in. Dia: M anti-∥, $\chi_m<0$, pushed out. Ferro: keeps M.
- Half-cyl coax $\mu_1|\mu_2$: B shared: $B = \frac{\mu_1\mu_2}{\mu_1+\mu_2}\frac{I}{\pi s}$.

21 · Electric ↔ magnetic swap

	P	M
uniform: where	caps (pierced)	sides (∥)
sphere inside	$-\mathbf{P}/3\epsilon_0$	$+\frac{2}{3}\mu_0\mathbf{M}$
long rod ∥	≈0	$\mu_0\mathbf{M}$
thin disk ⊥	$-\mathbf{P}/\epsilon_0$	≈0
auxiliary	D: ρ_f	H: I_f
fixed free src + material	E ÷ εr	B × μr

23 · Exam from 2 semesters ago (student report)

- Lots of **D, E, P**: chain §1, shortcuts §2, interfaces §12b.
- **Cylinder with off-axis hole:** currents from area ratio §14, then superpose.
- **A you can't integrate:** guess direction (∥ J), write the one curl component, solve the ODE §15; check $\nabla\cdot\mathbf{A} = 0$, continuity at boundaries.

22 · Curl/div one-liners (cyl)

$F(s)\hat{\phi}$: div 0, curl $\frac{1}{s}(sF)'\hat{z}$. $F(s)\hat{z}$: div 0, curl $-F'\hat{\phi}$. $F(z)\hat{x}$: curl $F'\hat{y}$. Valid B needs div = 0. $\hat{\phi}/s$: curl hides line current on axis.